Eur J Clin Pharmacol (2003) 58: 747–755
DOI 10.1007/s00228-002-0553-3

PHARMACOKINETICS AND DISPOSITION

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Intravenously administered digoxin in patients with acute atrial fibrillation: a population pharmacokinetic/pharmacodynamic analysis based on the Digitalis in Acute Atrial Fibrillation trial

Received: 16 April 2002 / Accepted: 4 December 2002 / Published online: 19 February 2003
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Abstract *Objective*: Atrial fibrillation is commonly treated with intravenously administered digoxin. The main objective of this study was to investigate the relationship between plasma concentration of digoxin and heart rate.
Subjects and methods: Plasma concentrations of digoxin were analysed in 105 patients allocated to digoxin therapy in the Digitalis in Acute Atrial Fibrillation (DAAF) trial. A pharmacokinetic/pharmacodynamic (PK/PD) model for the relationship among digoxin dose, plasma concentration and heart rate in patients remaining in atrial fibrillation was constructed using non-linear, mixed-effect modelling. One hundred and twenty-two placebo-treated patients were included as a control group. In 56 patients, one late sample at 16 h after the first dose of digoxin was obtained while in 49 patients an early sample at 0.25–0.5 h and a late sample 16 h after the first dose were obtained. Heart rate was measured at 0, 2, 6, 12 and 16 h after inclusion, with data from 98, 89, 67, 56 and 53 patients available at each time point, respectively.
Results: A two-compartment model best described the time course of digoxin concentrations in plasma. Digoxin and creatinine clearance correlated strongly and mean plasma concentration of digoxin at 16 h was within recommended levels (1.6 ± 1.0 nM). The decrease in heart rate in placebo-treated patients was, on average, 0.5 beats/min (bpm) per hour. In patients on digoxin, a linear relationship between the estimated digoxin concentration at the effect site and the drop-in heart rate was found. The half-life for the digoxin distribution to the effect compartment was approximately 3.8 h. The degree of reduction was related to the initial heart rate and patients with higher heart rate had a more pronounced decrease. The model predicted that a digoxin concentration of 1 nM at the effect site reduces heart rate by 9.4%.
Conclusion: A PK/PD model for the relationship between the plasma concentration of digoxin, the estimated concentration at the effect site and the reduction in heart rate during atrial fibrillation could be defined using a population pharmacokinetic approach. Our data indicate that a more aggressive dosing regimen of digoxin may be more effective in terms of heart rate reduction.

Keywords Atrial fibrillation · Digoxin · Population PK/PD modelling

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Introduction

Atrial fibrillation is a common medical problem. The incidence increases with age, and the prevalence is approximately 4% over 60 years and 10% over 75 years of age [1, 2, 3, 4, 5, 6]. The haemodynamic situation is characterised by increased atrial filling pressures, and the patient may be haemodynamically compromised.

Treatment of patients with acute atrial fibrillation aims at a reduction in heart rate and conversion to sinus rhythm. The standard treatment is pharmacological or DC electrical cardioversion. Treatment with intravenously administered digoxin is still the first choice in many hospitals. However, the scientific documentation in support of a positive effect on conversion to sinus rhythm is absent. Three randomised placebo-controlled studies have been reported [7, 8, 9], and none of them

showed any effect on conversion to sinus rhythm, relative to placebo. The tolerability to digoxin was, however, good in the three trials. The reason for the frequent use of digoxin in this situation may be the theoretical advantage of drug, over other anti-arrhythmic drugs, as it combines a positive inotropic effect with a decreasing effect on ventricular rate.

The mechanisms of action of digitalis [10, 11, 12, 13, 14, 15, 16] and its pharmacokinetic (PK) properties [17, 18, 19, 20, 21, 22, 23] have been described previously. Several investigators have also described the relationship between serum concentration of digoxin and cardiac effects in animals [24], healthy individuals [25, 26, 27] and patients with atrial fibrillation [28, 29, 30, 31, 32, 33] or through computer-based simulation [34]. The development of population pharmacokinetic methods has increased the possibility to describe and investigate the pharmacokinetic/pharmacodynamic (PK/PD) relationship for drugs in patients in which only a few samples for drug analysis can be taken [35, 36]. Such methods have also been used to study digoxin [37, 38, 39, 40, 41, 42]. In 1974, Jeliffe et al. proposed a standardised "Nomogram for digoxin therapy" for oral loading and maintenance therapy with digoxin [43], but no international consensus on a "gold standard" regimen for the initiation of digoxin therapy in acute situations exists. The Digitalis in Acute Atrial Fibrillation (DAAF) trial was performed in order to study the effects of intravenously administered digoxin on heart rate and the probability of conversion to sinus rhythm in patients with acute atrial fibrillation. The main efficacy results have been reported elsewhere [8].

The present study aims at describing the response of heart rate in relation to the achieved plasma concentrations of digoxin, using a population PK/PD approach.

Subjects and methods

The present report is a post-hoc analysis based on the data from patients recruited in the DAAF trial [8]. This trial was performed in 13 Swedish hospitals and was a randomised, double-blind comparison of intravenously administered digoxin and placebo in patients with acute atrial fibrillation. The primary end point was conversion to sinus rhythm within 16 h after randomisation, while effects on heart rate and safety were secondary end points. The DAAF trial provided an opportunity to describe dose-concentration–response in a patient population receiving i.v. digoxin, through the building of a population PK/PD model.

Patients over 18 years of age presenting with atrial fibrillation with a maximal duration of 7 days were eligible. Main exclusion criteria were ongoing treatment with digitalis or contraindications to this drug, treatment with anti-arrhythmic drugs, haemodynamic instability, heart rate under 60 beats per minute (bpm) or over 170 bpm, serum potassium under 3.3 mmol/l or serum creatinine above 300 μmol/l.

Each patient gave informed consent before inclusion. A complete medical history, physical examination and a standard 12-lead electrocardiogram (ECG) to confirm the diagnosis of atrial fibrillation were performed. Blood samples were drawn for serum levels of potassium and creatinine in all patients. The patients were under telemetric monitoring or close clinical supervision during the trial.

Patients were randomly allocated to double-blind treatment with intravenously administered bolus injections of digoxin (Lanacrist 0.25 mg/ml, Draco Läkemedel AB, Lund, Sweden) according to a predefined schedule based on body weight or placebo. The initial digoxin dose was 0.25 mg in patients below 50 kg and 0.50 mg in patients above 50 kg body weight. At 2 h and 6 h after the initial dose, a further 0.25 mg was administered to patients between 40 kg and 75 kg weight. In patients above 75 kg, the second and third doses could be increased to 0.5 mg at the discretion of the responsible investigator. The protocol allowed an additional dose of 0.25–0.50 mg study drug during the 16-h trial period, if clinically indicated. The aim of the regimen was to administer a total digoxin dose of 0.015–0.020 mg/kg body weight. For unstable patients in clinical need of additional anti-arrhythmic therapy, the choice among extra digoxin/placebo, other antiarrhythmic drugs or DC conversion was allowed at the discretion of the local investigator.

A standard 12-lead ECG was recorded before entry and at the predefined intervals 2, 6, 12 and 16 h after the first injection, and when conversion to sinus rhythm occurred. To enable computerised analysis, the ECGs collected 0, 2, 6, 12 and 16 h after inclusion were digitised manually by one person (B.H.). Measurements were done in the lead with the largest R-wave amplitude, and the highest point of all consecutive R-waves on the respective ECG at any time point was indicated after the software computed mean heart rate. An ECG had to fulfil the following criteria to be included in the digitising process; a steady isoelectric level with well-defined ECG complexes and a correct calibration for paper speed and voltage amplitude. The numbers of complexes utilised were 44.6 ± 12.9 at baseline, 44.3 ± 11.2 at 2 h, 40.4 ± 13.3 at 6 h, 39.4 ± 14.4 at 12 h and 42.7 ± 14.4 at 16 h after inclusion.

The study was initiated and monitored from Sahlgrenska University Hospital/Östra (SU/Ö), Gothenburg, Sweden. The study was approved by the Swedish Medical Products Agency and local ethics committees.

Bioanalytical assay

The plasma digoxin concentrations were determined at the Therapeutic Drug Monitoring laboratory at Huddinge Hospital using the Abbott TDX immunoassay. Limit of quantification was set at 0.4 nM (1 mg/l = 1.3 nM) and the within-assay coefficient of variation and the day-to-day variation were both below 10% levels equal to the limit of quantification.

Pharmacokinetic methods

The intention was to obtain an early plasma sample at 0.25–0.5 h and one late sample 16 h after administration of the first dose for determination of the plasma concentration of digoxin. A population kinetic approach using mixed-effect modelling was required, as the sparse concentration data did not allow individual kinetic modelling. The NONMEM software was chosen for the evaluation [44].

Briefly, all plasma concentration data were pooled in the analysis but the identity of each data point was retained with regard to subject characteristics and individual doses administered. A structural PK model was selected that best described the typical time-course of plasma concentrations. A covariate model was built that related subject-specific factors, e.g. body weight or renal function, to the parameters of the structural PK model. Lastly, a statistical model was defined that described the between-subject variability in the parameters, which was modelled using exponential distributions and the residual error; i.e. the remaining deviations between the observed and the model-predicted values. These deviations, which may be due to, for example, a suboptimal structural model, measurement errors and random changes in PK parameters over time, were modelled using both additive and proportional components, where either component was omitted if redundant. The results are presented as the typical population values for the PK parameters and their between-subject variability. The covariate relationships are expressed as regression functions.

Individual estimates of PK parameters are provided by NONMEM, based on a Bayesian approach [36] considering both the population distribution of PK parameters and the individual data. Specifically for this evaluation, one- and two-compartment structural PK models were tested. The covariate model was built in a stepwise manner testing the factors gender, age, body weight and estimated creatinine clearance on the PK parameters. Renal function was estimated for each patient using the Cockcroft-Gault formula [45]. Estimates of creatinine clearance exceeding 140 ml/min were set at 140 ml/min as an overprediction of this parameter and may occur in patients with a low age and a high body weight.

NONMEM version V, level 1.0 – with the subroutines ADVAN3 and TRANS4, i.e. parametrisation in terms of clearance (CL), intercompartmental clearance (Q), central (Vc) and peripheral (Vp) volumes of distribution – was used for the final PK model. The subroutines ADVAN5 and TRANS1 were used when determining the PD parameters. The first-order (FO) method was used for both the PK and PD analyses.

Evaluation of the PK/PD relationship

Two types of response were followed in the main efficacy study: first, the change in the ventricular heart rate while the patients still were in atrial fibrillation and, second, the possible conversion to sinus rhythm [8]. Due to the high frequency of spontaneous conversions to sinus rhythm in the placebo group and the lack of a statistical difference in conversion rate between placebo and active treatment, it was judged that PK/PD modelling for this variable would not be successful. The modelling in the present substudy therefore focused on the ventricular heart rate in patients with ongoing atrial fibrillation. A link model assuming a non-instantaneous distribution to the effect compartment was applied due to the well-known delay in response after intravenous administration of digoxin. Both linear and E_{max} models for the relationship between drug concentration at the effect site and the reduction in heart rate were tested. The change in heart rate among the placebo patients was considered in the model. The PK/PD relationship with regard to heart rate was evaluated in a two-step procedure. The individual PK parameters were first used to calculate the individual concentration–time relationship for digoxin, which in turn was used as a basis for determining the PD parameters. Previous history of congestive heart failure and presence of heart failure at inclusion were included as categorical covariates in the initial PK/PD model.

Statistics

Continuous parameters are described as mean values ± 1SD, with the range within brackets, when appropriate. For group comparison of demographics, a two-tailed *t*-test was used for continuous variables, while the Chi-square test was used for discrete variables. For these analyses, a P value ≤ 0.05 was considered statistically significant.

In the population analyses, discrimination between rival, hierarchical models were based on the differences in the objective function value produced by NONMEM. This value is minus twice the log likelihood (−2× log likelihood), and such differences are approximately Chi-square distributed and can therefore be used in hypothesis testing. A P value of 0.01 was required to accept a more complex model over a simpler one. In addition, for acceptance of a model, adequate goodness-of-fit, as evaluated by residual analyses, was required.

Results

Population

In total 117 patients were randomised to active treatment with digoxin in the DAAF trial. Six patients never received the drug due to early conversion to sinus rhythm and in a further six patients no sample was taken for drug analysis or the concentrations measured were below the limit of quantification in the analytical assay. Therefore, 105 of the patients were included in the pharmacokinetic evaluation. In three of these patients, the Cockcroft-Gault formula predicted creatinine clearance values above 140 ml/min, and they were assigned the value 140 ml/min. Descriptive statistics for the demographic variables are given in Table 1. All 122 patients allocated to placebo therapy in the DAAF trial were included as control group. There were no significant differences relative to digoxin-treated patients regarding baseline parameters or demographics [8].

In 49 patients two samples (early and late) were obtained and in the remaining 56 only the late sample was. No demographic variables differed significantly between these groups. Of the patients, 105 received a first dose of digoxin, 88 a second dose and 63 a third dose. The mean doses administered were 0.48 mg at inclusion, 0.31 mg after 2 h and 0.32 mg at 6 h, while four patients received an additional fourth dose of 0.25 mg. The distribution of doses administered are shown in Table 2.

Plasma concentrations of digoxin in the early sample at 0.35 ± 0.06 h (0.25–0.5 h) after inclusion were 9.6 ± 3.2 nM (3.6–17.6 nM), while the late sample at 16.1 ± 0.37 h (15.6–18.6 h) showed a concentration of 1.6 ± 1.0 nM (0.39–7.9 nM). The average total dose given was slightly higher among males than among females, on average 1.0 mg and 0.87 mg, respectively.

Pharmacokinetics

A two-compartment, open model best described the time course of digoxin concentrations. A one-compartment model resulted in a significantly worse fit as judged by a 44 unit higher objective function value compared with the two-compartment model.

For the central (V_c) and peripheral (V_P) volumes of distribution, as well as for the intercompartmental clearance (Q), only the population average values were estimated and not the between-patient variability or any covariate relationships. The reason was that since approximately 50% of the patients only had the late

Table 1 Descriptive statistics of the study population receiving active treatment with digoxin

Demographics	
Age (years)	67 ± 12 (21–89)
Weight (kg)	77 ± 15 (47–139)
Creatinine clearance[a] (ml/min)	71 ± 27 (20–140)
Heart rate at inclusion (bpm)	120 ± 22 (74–173)
Medication at inclusion (%)	
Beta-blockers	23
Sotalol	15
Rate-limiting calcium blockers	9
Angiotensin converting enzyme inhibitors	17

[a]Estimated according to the Cockcroft-Gault formula

Table 2 Distribution of doses administered in the study population. The number of patients for each dose level at each occasion is presented

	0.5 mg	0.25 mg	0.125 mg	Total *n*
Dose 1	96	9		105
Dose 2	23	64	1	88
Dose 3	18	45		63
Dose 4		4		4

sample taken for digoxin concentration they contributed with limited individual information on these parameters. It was also found that the drop in objective function for the population model was negligible when these parameters were allowed to vary and that by including the covariate body weight on V did not improve the fit. Using individual estimates of CL from the basic model, a clear correlation (r = 0.61) was identified with CrCL, an introduction of CrCL as a linear relationship with CL decreased the objective function with 26 units. No other covariate was found to correlate with the individual CL estimates. The estimated relationship with CrCL was as large as was allowed. A stronger relationship would have resulted in population predictions of negative CL values at CrCL below 20 ml/min. The population parameter estimates of the final PK model for digoxin clearance are presented in Table 3 and goodness-of-fit in Fig. 1.

Relationship between digoxin concentrations and heart rate

Heart rate rapidly decreased in digoxin-treated patients who remained in atrial fibrillation (Table 4). The decrease was best described as proportional to the baseline rate and with a linear relationship with the estimated digoxin concentration at the effect site. No factors influencing the response, including previous history of congestive heart failure and/or signs of heart failure at inclusion, were identified. The relationship between the individually predicted and observed heart rate is presented in Fig. 2. The final model is: heart rate = baseline heart rate$\times(1+\theta_{placebo}\times time)\times(1+\theta_{digoxin}\times Ce)$, where Ce is the hypothetical digoxin concentration at the effect site, $\theta_{placebo}$. The value of the equilibration rate constant, ke_0, indicates that the half-life for the distribution of digoxin to the effect compartment was approximately 3.8 h. The model predicts that for a subject with a baseline heart rate of 125 bpm, the placebo response is on average 0.5 bpm per hour. In the same subject, a digoxin concentration of 1 nM at the effect site reduces the heart rate by 9.4%. The residual error for the model is approximately 6%, indicating that all important factors were identified.

Figure 3 shows the typical concentration–time profile in plasma for digoxin, with the most common dosage regimen used in the DAAF trial, which was 0.5, 0.25 and 0.25 mg digoxin given at 0, 2 and 6 h, respectively. In the same figure, the estimated profile in the effect compartment, scaled with a factor of 10 for readability, is shown. Effects on heart rate for two hypothetical patients with baseline rate of 100 bpm and 150 bpm, respectively, are also shown. The estimated effects and concentration profile in plasma and effect compartments with the dosage regimen 0.5, 0.5 and 0.25 mg digoxin given at 0, 1 and 4 h, respectively, after inclusion are shown in Fig. 4. With the latter regimen, the concentration in the effect compartment increases more rapidly resulting in a more rapid response and a more pronounced maximum effect.

Discussion

Patients with acute atrial fibrillation are common at emergency wards. Treatment with intravenously

Table 3 Population parameter estimates for final pharmacokinetic (PK) and pharmacodynamic (PD) models

	Parameter	Unit	Typical value	RSE[a]	Between-subject variability (%)
PK model	CL[b]	l/h	9.88	0.15	66
	θ_{CrCL}[b]	min/ml	0.02	N.E.[d]	N.E.[e]
	Vc	l	27.8	0.25	N.E.
	Q	l/h	71.8	0.076	N.E.
	Vp	l	444	0.07	N.E.
	Residual error	nM	0.24	0.11	N.E.
PD model	Baseline[c]	bpm	119	0.013	17
	$\Theta_{placebo}$[c]	1/h	−0.0040	0.28	230
	$\Theta_{digoxin}$[c]	1/nM	−0.0939	0.10	N.E.
	k_{eo}	1/h	0.184	0.15	N.E.
	Residual error	%	9.0	46.6	N.E.

[a]RSE, relative standard error, is the SE divided by the parameter estimate
[b]CL for the typical patient is described by the equation: $CL = CL\times(1+\theta_{CrCL,}\times(CrCL-70))$
[c]HR for the typical patient is described by is described by the equation: $HR = baseline\times(1+\theta_{placebo,}\times time)\times(1+\theta_{digoxin}\times Ce)$, where Ce is the effect compartment concentration
[d]Not estimated. Estimate on the boundary of physiological range
[e]Not estimated. Ventricular heart rate according to ECG measurements at different time points in relation to first dose administration. Only data from patients having ongoing atrial fibrillation at each time point are included

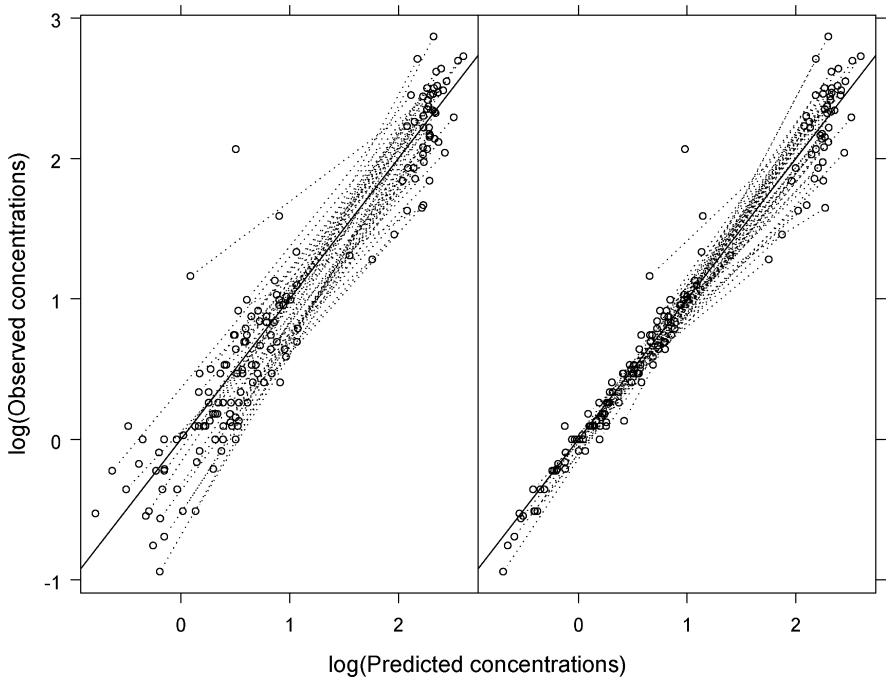

Fig. 1 Observed versus predicted concentration. Predictions are made based on population (*left panel*) or individual (*right panel*) parameters

Table 4 Ventricular heart rate according to electrocardiogram measurements at different time points in relation to first dose administration. Only data from patients having ongoing atrial fibrillation at each time point are included

Time point (h)	Digoxin		Placebo	
	n	Heart rate (bpm)	*n*	Heart rate (bpm)
0	98	120 ± 22 (74–173)	116	123 ± 23 (62–180)
2	89	104 ± 21 (60–160)	98	117 ± 22 (60–160)
6	67	92 ± 19 (56–127)	84	110 ± 24 (64–157)
12	56	89 ± 19 (47–140)	69	110 ± 22 (58–160)
16	53	91 ± 20 (51–138)	59	116 ± 25 (54–171)

administered digoxin is still a common choice in many hospitals despite the lack of scientific documentation supporting an effect on conversion to sinus rhythm [6, 7, 8]. Reasons for the frequent use of digoxin may be the theoretical advantage of a positive inotropic effect and a modifying effect on the ventricular rate. One further explanation may be that digoxin is well tolerated if administered in recommended doses, while most other anti-arrhythmic drugs are associated with a higher risk of complications, such as worsened heart failure or malignant arrhythmias [46, 47, 48, 49]. Digitalis glycosides have effects on both the mechanical and electrical properties of the heart [10, 11, 12, 13, 14, 15, 16, 19, 22]. These effects are directly pharmacological on the myocytes and also mediated through the autonomic nervous system. Furthermore, there is a difference in the autonomic effects of digoxin, so that the vagal effects are predominant with low doses, while increased sympathetic activity, with risk for proarrhythmias, are predominant at high doses.

For emergency purposes, digoxin is administered parentally. The recommended dosing schedules differ somewhat between countries. In Sweden, the speed of injection is recommended to be maximally 0.1 mg/min. The bolus dose of digoxin for a normal person (70 kg of weight) is 0.5–0.75 mg. This can be followed by 0.25 mg i.v. every 3 h, up to a total maximal dose of 1.5 mg. Thereafter, oral maintenance therapy is instituted. The dosing schedule in the DAAF trial was somewhat more aggressive, aiming at administration of a total digoxin dose of 0.015–0.020 mg/kg body weight to achieve therapeutic plasma levels of digoxin 16 h after inclusion. Pharmacokinetics of digoxin in humans has been evaluated in several studies [17, 18, 20, 22, 25, 26, 27]. Most commonly, two- or three-compartment models have been applied to describe the concentration–time profile in plasma. Distribution to skeletal muscle and heart is extensive. However, the distribution within the heart is not uniform dependent on the presence of myocardial scarring, fibrosis or ischaemia [50, 51, 52, 53]. According to some investigators, it is also dependent on heart rate and rhythm [54]. Digoxin is partly excreted unchanged through glomerular filtration. Digoxin clearance has in many studies shown correlation with estimated creatinine clearance, even if tubular secretion also plays a significant role.

In the present study, only one or two plasma concentrations of digoxin were determined in each patient.

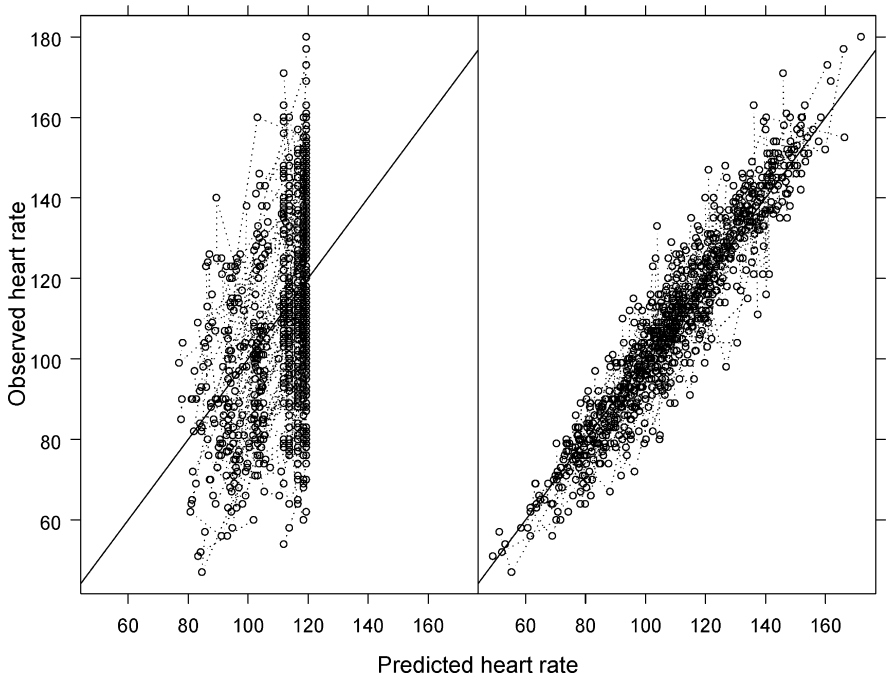

Fig. 2 Observed versus predicted heart rate among patients with ongoing atrial fibrillation receiving active treatment with digoxin. Predictions are made based on population (*left panel*) or individual (*right panel*) parameters

Despite the limited amount of data a two- compartment structural model could be identified. The population PK parameters were in agreement with data in studies with several analyses of plasma concentrations. The relationship between estimated creatinine clearance and digoxin clearance is in accordance with previous results. In previous studies of the concentration–effect relationship for digoxin, different effect measures have been applied such as the QS_2 index [25, 26, 29, 32], the left ventricular pressure (dP/dt) [53] and the left ventricular ejection time (LVETc) and pre-ejection period (PEPc) [56]. It is well recognised that there is a delay in the distribution of digoxin from plasma into the myocardium [55] and no direct temporal relationship between plasma drug concentrations and response is therefore expected.

A general problem when studying patients with acute atrial fibrillation is the high proportion of dropouts due to conversion to sinus rhythm, which may cause bias when using conventional modelling techniques. Using mixed-effect modelling, these problems are taken into consideration. We found a close relationship with the heart rate during atrial fibrillation and could define a robust model, based on the low relative standard error of the estimates for the various parameters and the low residual error magnitude, by which the outcome of different dosage regimens can be simulated. Our results corroborate that there is no direct temporal correlation between the plasma concentration of digoxin and effects on heart rate, while a close correlation with the estimated concentration at the effect site in the myocardium was present. A linear function adequately described this relationship, even if an E_{max} model was expected to be more appropriate. Our interpretation is that the range of concentrations in this study was too narrow to describe the non-linear parts of the concentration/response curve.

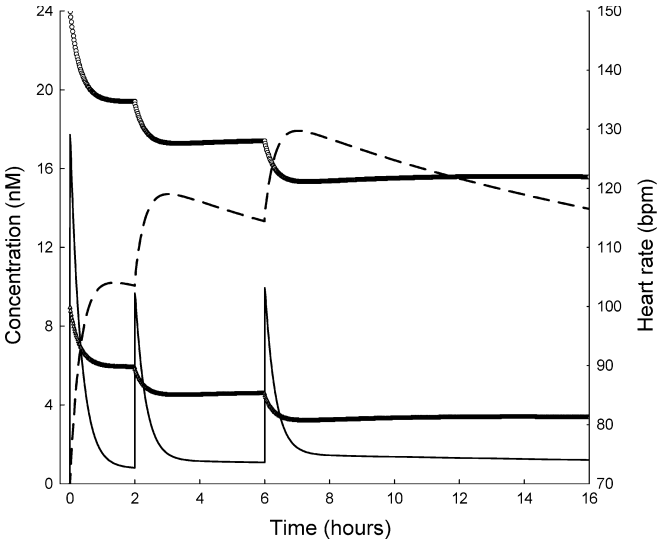

Fig. 3 Typical concentration–time profiles for digoxin in plasma (*continuous line*) and in effect compartment (*broken line*, scaled upwards with a factor of 10) with bolus doses 0.5, 0.25 and 0.25 mg given i.v. at 0, 2 and 6 h. Based on average pharmacokinetic parameters in the present study. Predicted heart rate with this regimen assuming baseline value 150 bpm (*open circles*) or 100 bpm (*open triangles*) and no conversion to sinus rhythm also shown

Digoxin decreased heart rate in patients remaining in atrial fibrillation. The model predicts that following each dose, maximal heart rate reduction occurs after about an hour (Fig. 3). This is much shorter than the equilibration half-life for the effect compartment, which can be explained by the rapid decrease in the plasma concen-

trations following the digoxin dose. A linear relationship between the estimated digoxin concentration at the effect site and effects on heart rate was found. Noteworthy is that this effect was proportional so those patients with a high heart rate had a more pronounced decrease than patients with a lower heart rate.

In 1975, Goodman et al. presented a study on effects of acute i.v. administration of digoxin to a group of humans with intact cardiac innervation and a group who had undergone cardiac transplantation and thereby had completely denervated hearts [57]. They found the electrophysiological effects by digoxin on AV conduction to be dependent on cardiac innervation and not on a direct pharmacological effect per se. Our findings on the effects of digoxin on heart rate are in accordance with this and are probably explained by the vagotonic effects of the drug on AV conduction, since a high heart rate is due to a high sympathetic, or a low vagal, nerve tone, whereas the increase in vagal tone caused by digoxin will result in a decrease in heart rate proportional to the baseline heart rate.

In previous placebo-controlled studies, the effects of digitalis on heart rate have been studied. Falk et al. [7] could not demonstrate any clinically relevant effect on heart rate, while in the DAAF trial [8] and in the trial conducted by Jordaens et al. [9] digoxin decreased heart rate significantly relative to placebo after 2 h and after 30 min, respectively. One possible explanation for these discrepancies is the differences in dosing regimen used. In the study by Falk et al. [7], 36 patients were randomised to orally administered capsules with 0.6 mg, 0.4 mg, 0.2 mg and 0.2 mg digoxin ($n=18$) or placebo ($n=18$) at times 0, 4, 8 and 14 h after inclusion. In both the Joardens trial [9] and the DAAF trial [8], digoxin or placebo was administered as i.v. bolus injections or infusions. Joardens et al. [9] randomised 39 patients among treatment with 0.75 mg, 0.25 mg and 0.25 mg digoxin ($n=19$) or corresponding placebo ($n=20$) at times 0, 4 and 12 h after inclusion, while in the DAAF trial [8] 239 patients were randomised, after a weight-adjusted dosing schedule, among 0.25–0.5 mg, 0.25–0.5 mg and 0.25–0.5 mg digoxin ($n=117$) or placebo ($n=122$) at times 0, 2 and 6 h after inclusion. One further factor may be differences in arrhythmia duration. This is an important factor regarding probability for spontaneous conversion to sinus rhythm [58] and, according to our data, also for heart rate at inclusion.

Patients with overt or past clinical congestive heart failure responded with a heart rate reduction of the same magnitude as other patients. One may have expected the sympathetic activation in these patients to prevent this. Obviously the combined effect of digoxin induced vagal stimulation, the positive inotropic effect and possibly a sympatho-inhibitory response [59] was sufficient to counteract this. As haemodynamic instability was an exclusion criterion in the DAAF trial, our data do not include patients with severe ongoing haemodynamic instability.

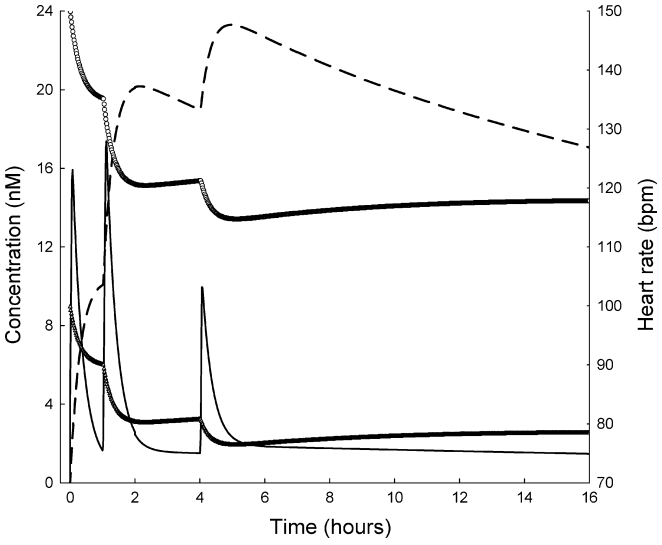

Fig. 4 Same as Fig. 3 but with bolus doses 0.5, 0.5 and 0.25 mg given i.v. at 0, 1 and 4 h. Predicted heart rate with this regimen assuming baseline value 150 bpm (*open circles*) or 100 bpm (*open triangles*) and no conversion to sinus rhythm also shown

From our data and also from previous PK studies, it seems clear that as much drug as tolerable should be administered during the loading phase due to the slow build-up of concentrations at the effect site. The dosage regimen applied in the DAAF trial resulted in a relatively even concentration–time profile in the effect compartment from 3 h onwards. However, by administering a higher dose at the second dosing event and by shortening the dosage intervals, the effect on heart rate might have been even more substantial during the early phase, as exemplified in Fig. 4. The effects on heart rate are accomplished earlier, while the risks for side effects, such as bradyarrhythmias and nausea, are unknown but probably low.

Conclusion

A PK/PD model for the relationship among the plasma concentration of digoxin, the estimated concentration at the effect site and the reduction in heart rate during ongoing atrial fibrillation could be defined. The model shows that the effect on heart rate is already pronounced 1 h after an i.v. bolus dose, and the model can be used for simulation of the response over time with different dosage regimens. The data indicate that a more aggressive therapeutic regimen may be more effective in terms of heart rate reduction, where watchful observation of maximal effects on heart rate during approximately 1 h before decision on the next dose may be recommended. Further studies on such regimens are warranted.

Acknowledgements We thank Gunilla Norman, RN, and Eva Alfvegren, RN, Division of Cardiology, SU/Ö, and Gillis Johnsson, MD, PhD, AstraZeneca AB. Sources of financial support: Swedish Society of Cardiology, Göteborg Medical Society, AstraZeneca AB, Mölndal, Sweden. Support with drugs and logistics: AstraZeneca

AB, Mölndal, Sweden. Participating centres and investigators included: (1) Co-ordinating centre: SU/Ö, Göteborg. Björn Hornestam, MD; Peter Held, MD, PhD; Birgitta Bartholdsson, MD. (2) Trialists and centres (no. of included patients in parentheses): SU/Ö (38 patients) as above. Skellefteå Hospital (4 patients): Kirsti Kärki, MD; Kurt Boman, MD, PhD. Örnsköldsvik Hospital (2 patients): Olle Löwheim, MD. Falun Hospital (10 patients): Cecilia Dahlen, MD. Västerås Hospital (4 patients): Stellan Bandh, MD. Karlstad Hospital (4 patients): Christer Åbjörn, MD. Skövde Hospital (29 patients): Torbjörn Lundström, MD, PhD. Lidköping Hospital (34 patients): Magnus Peterson, MD. Uddevalla Hospital (26 patients): Björn W. Karlson, MD, PhD. Trollhättan Hospital (15 patients): Arne Redfors, MD, PhD; Lennart Sandstedt, MD; Magnus Wallin, MD. SU/Sahlgrenska, Göteborg (24 patients): Björn W. Karlson, MD, PhD. Mölndal Hospital (22 patients): Lennart Falk, MD. Varberg Hospital (29 patients): Tommy Carlsson, MD; Fredrik Scherstén, MD.

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